

Predicting treatment delays of more than 90 days in prostate cancer: added value of county rurality, county income and marital status over clinical characteristics

A machine learning cohort study of 330,827 men treated with surgery or radiotherapy in US SEER registries, 2010 to 2022

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Abstract

BACKGROUND It is unclear how much area and marital characteristics add to recorded clinical need in predicting delayed prostate cancer treatment.

METHODS In this retrospective cohort study of SEER 17 registry data, we included men aged 40 or over diagnosed 2010 to 2022 with localised or regional prostate cancer and recorded radical prostatectomy or radiotherapy. The outcome was first recorded treatment over 90 days after diagnosis. Penalised logistic regression and LightGBM were cross-fitted in ordered steps (year; clinical need; county rurality, county income and marital status; treatment type) within risk groups. The primary estimand was the out-of-fold log-loss skill added by the area and marital block.

RESULTS Of 330,827 men, 39.7% waited more than 90 days, rising from 36.0% in 2010 to 52.3% in 2022. Prediction was weak (AUC 0.592 to 0.663). The area and marital block added 0.62 percentage points of skill (95% interval 0.39 to 0.87) under logistic regression and 0.99 (0.73 to 1.30) under LightGBM, and was stable under tuning, fold-assignment and clustering checks. Standardised delay was 9.2 and 8.8 points lower in non-metropolitan counties not adjacent to a metropolitan area than in large metropolitan counties, and 6.7 and 5.9 points higher for never-married than married men (post hoc two-model rule; no intervals).

CONCLUSIONS Recorded information predicts little of who waits beyond 90 days. Area and marital characteristics add a small, repeatable increment. Delay was more common in large metropolitan, higher-income counties, which is not evidence of better care elsewhere because recorded treatment also differs by area.

Keywords prostate cancer; time to treatment; health services research; machine learning; health equity

KEY POINTS

Question How much do county rurality, county income and marital status add to recorded clinical need in predicting a wait of more than 90 days for radical prostatectomy or radiotherapy?

Findings In this cohort study of 330,827 US men, registry information predicted delay weakly (AUC 0.592 to 0.663). Area and marital characteristics added 0.62 and 0.99 percentage points of log-loss skill in all men under two model types, with 95% intervals above 0 in every risk group.

Meaning Who waits beyond 90 days is mostly not captured by registry data. Registries with referral dates, treating sector and comorbidity could show what the small area and marital signal represents.

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Data
SEER Research Data, 17 registries, November 2025 submission.

Code and protocol
github.com/tanmayhinge/seer-prostate

Supplementary material
Tables S1 to S11 and Figures S1 to S3.

Introduction

Time from diagnosis to first treatment is a widely used measure of the timeliness of cancer care. In the US National Cancer Database, the median time to first treatment for six early-stage cancers rose from 21 to 29 days between 2004 and 2013, and care at an academic centre was among the factors associated with longer intervals [1]. Intervals are longer for prostate cancer: the median was 79 days (interquartile range 55 to 117) in the same database for 2004 to 2015 [2], and the mean in SEER was 82.75 days in 2021 [3].

Multivariable studies describe who waits longer. In SEER-Medicare data, time to definitive treatment was longer for African American than for White men with localised prostate cancer, and longer in high-risk than in low-risk disease [4]. Across four cancers in SEER, lower-income and non-metropolitan patients had shorter intervals [5], and in the Tennessee cancer registry residence in rural Appalachian counties was associated with lower

odds of waiting more than 90 days for prostate cancer treatment (odds ratio 0.83, 95% CI 0.78 to 0.89) [6]. Care at an accredited hospital was associated with a longer median interval in low-income but not high-income counties [7]. Unadjusted comparisons can mislead: in SEER, unadjusted all-cause mortality was lower among patients with longer intervals [8], a pattern we read as consistent with faster treatment of more aggressive disease.

These studies estimate how individual factors are associated with waiting after adjustment. They do not show how much of the variation in waiting can be predicted from recorded clinical need, or how much area and marital characteristics add beyond it. That added predictive value has been measured for other prostate cancer outcomes [9, 10], but in a PubMed search screened by one reviewer we found no study that applied it to time to treatment within prostate cancer risk groups. An out-of-sample increment describes the size of the signal without assuming its direction, and

fitting a flexible model such as gradient boosting [11] alongside penalised regression shows whether an increment merely reflects an under-fitted clinical model.

We therefore measured, within clinical risk groups of men treated with radical prostatectomy or radiotherapy, how much county rurality, county income and marital status add to recorded clinical need in predicting a wait of more than 90 days, and expressed that contribution as standardised percentages and an income concentration index.

Methods

Data and cohort

We used the SEER Research Data, 17 registries, November 2025 submission, linked to county attributes [12]. A case listing selected by site (prostate) and year of diagnosis gave 793,214 tumour records. Exclusions were applied in a pre-specified order (Figure 1): first primary cancer; localised or regional Combined Summary Stage; not a death certificate or autopsy case; aged 40 or over; diagnosed 2010 to 2022; first course including radical prostatectomy (surgery codes 50, 70 or 80, or A500, A700 or A800 from 2023) or radiotherapy [13]; a recorded or top-coded interval; and an interval above 0 days. The SEER coding manual instructs that treatment is coded as not done when the physician opts for active surveillance or watchful waiting, so men managed this way are not in the cohort [13].

Outcome and risk groups

The outcome used SEER's time from diagnosis to treatment in days, which ends at the first treatment of any kind; hormone therapy, not recorded in this export, can therefore end the interval before surgery or radiotherapy. The primary outcome was a wait of more than 90 days, a threshold used in US registry research [6] that matches the Australian optimal care pathway recommendation that surgery or radiotherapy begin within 3 months of diagnosis [14]. Intervals top-coded at 731 days or more counted as delayed; thresholds of 60, 120 and 180 days were sensitivity analyses. Risk groups were high (regional stage, clinical Gleason score 8 to 10, or PSA above 20 ng/ml), unknown (otherwise, Gleason score or PSA missing), intermediate (Gleason score 7 or PSA 10 to 20 ng/ml) and low (Gleason score 6 or lower and PSA below 10 ng/ml). Summary stage can incorporate pathology for surgically treated men.

Ordered feature steps

Step 0 contained year of diagnosis and a 2020 indicator. Step 1 added clinical need: stage category, clinical Gleason score, log PSA (with missing and top-code indicators) and age band. Step 2 added the area and marital block: marital status (7 categories), county Rural-Urban Continuum Code (5 levels plus unknown) and county median household income (16 bands, entered as a rank). Two of these three variables describe the county rather than the man, and because the export has no registry identifier the block can also carry differences in registry practice. Step 3 added treatment type, entered last be-

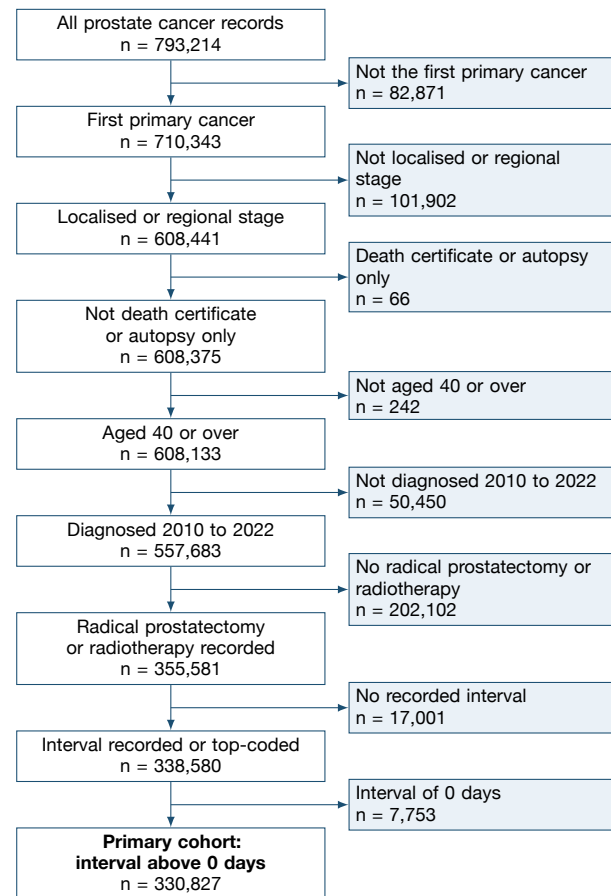


Figure 1. Inclusion flow for the primary cohort. Boxes give the number of records remaining after each step, in protocol order; shaded boxes give the number excluded at that step.

cause area and marital characteristics can influence the choice of treatment.

Models and validation

Penalised logistic regression used an L2 penalty, median imputation, standardisation and pairwise products of the clinical features [15]; LightGBM handled missing values internally [11]. All performance measures used out-of-fold predictions from 5-fold stratified cross-fitting within each risk group and in all men. Hyperparameters were tuned by 3-fold cross-validated log-loss on up to 50,000 men at step 3 and reused at every step. Log-loss skill is 1 minus a step's log-loss divided by that of the step 0 model; the primary estimand was the skill added from step 1 to step 2. We also report AUC and calibration slope. Because rurality and income are county measures and the export has no county identifier, intervals came from a cluster bootstrap over the 79 cells formed by rurality and income band (500 resamples of out-of-fold predictions, no refitting); they therefore do not include model-fitting variability.

Standardised percentages and income concentration

Standardised percentages came from g-computation with a step 2 model: each man kept his own clinical features and year while his area and marital features were set to a profile. The reference profile was married and

living in a county in a metropolitan area of 1 million or more, at the 83.33% quantile of county income rank. Because rurality and income are strongly correlated, area profiles set both together, at the median income band of men living in each type of area. After a development run showed the two model types disagreeing, we adopted a post hoc rule that a contrast is meaningful only when both models agree on 3 percentage points or more in the same direction, and, also post hoc, dropped the planned intervals for these contrasts. Income inequality in delay was summarised with the Erreygers-corrected concentration index [16], ranking men by county income with tied men sharing the average rank; the index is crude and does not separate income from rurality.

Robustness, selection and secondary outcome

Ten sensitivity scenarios each changed one setting (thresholds of 60, 120 and 180 days; risk groups from Gleason score and PSA only; prostatectomy without radiotherapy; excluding 2020; including 2023; including 0-day intervals; one primary only; excluding prostatectomy not otherwise specified). A later protocol amendment (version 1.9) added, before they were run, per-step tuning, 10 fold-assignment seeds, clustering by county income band alone, a stage-free clinical block, year as categories, and the concentration index by diagnosis period. For treated men without a recorded interval, the percentage delayed was bounded under extreme assumptions and reweighted by the inverse probability of having an interval. A secondary outcome was a record of radical prostatectomy or radiotherapy among intermediate- and high-risk men; SEER captures treatment incompletely (radiation therapy sensitivity 80% against Medicare claims, pooled across cancer sites) [17], so no record does not mean no treatment.

Protocol, reporting and disclosure

The protocol was committed to version control before the cohort was built and before any outcome model was fitted; 13 amendments are logged with reasons, and 3 made after seeing results are labelled as such. Commit times are recorded by the author's computer and the repository became public on 13 September 2026, after the analyses, so their order can be inspected but is not independently timestamped. The study was not registered. The analyses reported here are fixed at commit c1b6fd7 of the public repository. Reporting follows STROBE [18], RECORD [19] and TRIPOD+AI [20] (Supplementary Tables S10 and S11). The models are measurement tools and are not proposed for clinical use. As the SEER Research Data Use Agreement requires, statistics based on 1 to 4 men are suppressed. Analyses used Python 3.13, scikit-learn 1.9.1 and LightGBM 4.7.0 with seed 20260913.

Results

Cohort and delay

Of 793,214 records, 330,827 men met all criteria (Figure 1; Table 1). Rurality and county income overlapped heavily: 45.6% of men in non-metropolitan counties

not adjacent to a metropolitan area lived in the lowest county income quartile, against 1.1% of men in the largest metropolitan areas. The median wait was 77 days (interquartile range 52 to 116), and 131,326 men (39.7%) waited more than 90 days. Delay was more common in low-risk (47.9%) than intermediate-risk (42.8%) and high-risk men (32.8%), and rose from 36.0% in 2010 to 52.3% in 2022. Within every risk group, delay was more common in large metropolitan than in non-metropolitan counties not adjacent to a metropolitan area (low risk 50.7% against 39.0%; high risk 35.7% against 26.0%) and among never-married than married men (high risk 40.0% against 30.9%) (Supplementary Figure S1). For low-risk men, a longer wait is not necessarily worse care.

How much area and marital characteristics add

Prediction was weak: at step 2, AUC ranged from 0.592 to 0.663 across strata and models, and log-loss skill from 0.76% to 4.90% (Table 2). Clinical need added 3.01 percentage points of skill in all men under logistic regression (95% interval 2.69 to 3.38) and 3.53 in high-risk men, but only 0.14 (0.08 to 0.20) in low-risk men, whose clinical variables vary little by definition. The area and marital block added 0.62 points (0.39 to 0.87) under logistic regression and 0.99 (0.73 to 1.30) under LightGBM in all men, with every interval above 0 (Figure 2). In low-risk men it made up 82% and 98% of a small step 2 skill of 0.76 and 0.99 points. LightGBM had higher step 2 skill than logistic regression in every stratum, by 0.23 to 0.90 points (no interval computed), and calibration slopes were 0.98 to 1.00 for logistic regression and 0.88 to 1.03 for LightGBM.

Standardised differences and income concentration

Standardised delay was lower in non-metropolitan counties not adjacent to a metropolitan area than in large metropolitan counties, each at its typical county income: by 9.2 points under logistic regression and 8.8 under LightGBM in all men, and by 7.2 to 14.2 points across risk groups (Figure 3a). This contrast moves rurality and income together; changing rurality alone gave -8.7 and $+1.7$ points and changing income alone -0.9 and -8.7 points, so their separate contributions cannot be identified. Never-married men had standardised delay 6.7 and 5.9 points higher than married men, with agreement in every stratum except low risk (3.6 and 2.4; Supplementary Table S6). Setting all men to the reference profile changed the pooled percentage by only 1.1 and 1.6 points. Crude delay was concentrated in higher-income counties (Erreygers index 0.062, 95% interval 0.039 to 0.083; Figure 3b), but less so within diagnosis periods (0.059, 0.037 and 0.027 for 2010 to 2014, 2015 to 2019 and 2020 to 2022; Supplementary Table S7), while the share of men in the highest income quartile rose from 26.2% to 41.0%.

Robustness, selection and secondary outcome

Supplementary Tables S3 to S5, S8 and S9 give the full results summarised here. Tuning at each step left the in-

Table 1. Characteristics of the 330,827 men in the cohort, by county rurality

Characteristic	Overall	Metro, 1 million or more	Metro, 250,000 to 1 million	Metro, under 250,000	Nonmetro, adjacent to metro	Nonmetro, not adjacent to metro
Men, n	330,827	194,681	72,150	26,092	23,360	14,385
<i>Risk group</i>						
Low	17.4	17.6	16.5	18.3	17.1	17.5
Intermediate	38.6	38.6	39.0	38.6	37.8	37.2
High	39.6	38.9	40.6	38.8	41.0	41.8
Unknown	4.5	4.9	3.8	4.3	4.1	3.5
<i>Summary stage</i>						
Localised	77.5	77.7	76.7	78.7	77.4	77.9
Regional, direct extension	18.7	18.4	19.6	18.2	19.3	18.8
Regional, lymph nodes	3.7	3.9	3.7	3.1	3.2	3.3
<i>Marital status</i>						
Married	70.3	69.6	71.7	69.5	71.5	71.7
Never married	11.1	11.6	10.8	10.5	10.1	<5
Divorced	6.7	6.3	6.8	7.4	7.6	7.8
Widowed	3.0	2.7	3.1	3.5	3.7	3.8
Unknown	7.8	8.5	6.4	7.9	6.2	6.8
<i>County income quartile</i>						
Q1 (lowest)	8.0	1.1	5.0	20.2	37.7	45.6
Q2	24.1	12.4	32.5	58.4	45.1	44.4
Q3	35.1	42.7	33.9	18.0	11.0	8.9
Q4 (highest)	32.8	43.7	28.6	3.3	6.2	1.1
<i>First-course treatment</i>						
Radical prostatectomy	48.2	48.3	48.9	49.0	45.7	46.4
Radiotherapy	47.9	47.9	46.9	47.2	50.8	50.1
Both	3.9	3.8	4.2	3.8	3.5	3.5
<i>Days to first recorded treatment</i>						
Median	77	81	76	71	70	68
Interquartile range	52 to 116	55 to 121	50 to 112	48 to 106	46 to 105	43 to 103
Waited more than 90 days	39.7	42.5	37.6	34.2	33.2	32.2

Values are the percentage of men in the column unless stated. Rurality is the county Rural-Urban Continuum Code at diagnosis; 159 men with unknown rurality are included in Overall only. Separated (0.8%) and unmarried or domestic partner (0.4%) men are not shown. <5: suppressed under the SEER Research Data Use Agreement, either because it rests on 1 to 4 men or because a total would reveal such a cell. County income quartiles are formed from 16 bands of county median household income.

Table 2. Predictive performance and skill added by clinical need and by area and marital characteristics, by risk group and model

Risk group	Model	Men	Skill at step 2 (%)	AUC at step 2	Clinical need added (95% interval)	Area and marital added (95% interval)	Fold-seed range ^a
All men	Logistic	330,827	3.63	0.638	3.01 (2.69 to 3.38)	0.62 (0.39 to 0.87)	0.61 to 0.62
	LightGBM	330,827	4.17	0.648	3.18 (2.85 to 3.54)	0.99 (0.73 to 1.30)	0.98 to 0.99
Low	Logistic	57,495	0.76	0.592	0.14 (0.08 to 0.20)	0.62 (0.32 to 0.94)	0.59 to 0.63
	LightGBM	57,495	0.99	0.604	0.02 (−0.09 to 0.13)	0.96 (0.67 to 1.28)	0.95 to 1.01
Intermediate	Logistic	127,622	1.06	0.605	0.49 (0.41 to 0.56)	0.56 (0.35 to 0.83)	0.54 to 0.56
	LightGBM	127,622	1.48	0.615	0.55 (0.44 to 0.65)	0.93 (0.65 to 1.27)	0.91 to 0.97
High	Logistic	130,893	4.39	0.654	3.53 (3.16 to 3.86)	0.86 (0.60 to 1.15)	0.85 to 0.86
	LightGBM	130,893	4.90	0.663	3.72 (3.34 to 4.06)	1.18 (0.90 to 1.56)	1.16 to 1.20
Unknown	Logistic	14,817	1.07	0.610	0.65 (0.41 to 0.89)	0.42 (0.08 to 0.80)	0.32 to 0.48
	LightGBM	14,817	1.96	0.629	0.59 (0.27 to 0.95)	1.37 (0.80 to 2.19)	1.22 to 1.43

Skill is the percentage reduction in out-of-fold log-loss against the step 0 model (year of diagnosis only); added values are in percentage points. Intervals are 95% cluster bootstrap intervals over 79 rurality by county income cells (500 resamples of out-of-fold predictions, no refitting), so they exclude model-fitting variability.

^aRange of the area and marital increment over 10 fold assignments, from the robustness analysis with per-step tuning (protocol amendment 1.9). AUC: 0.5 is chance.

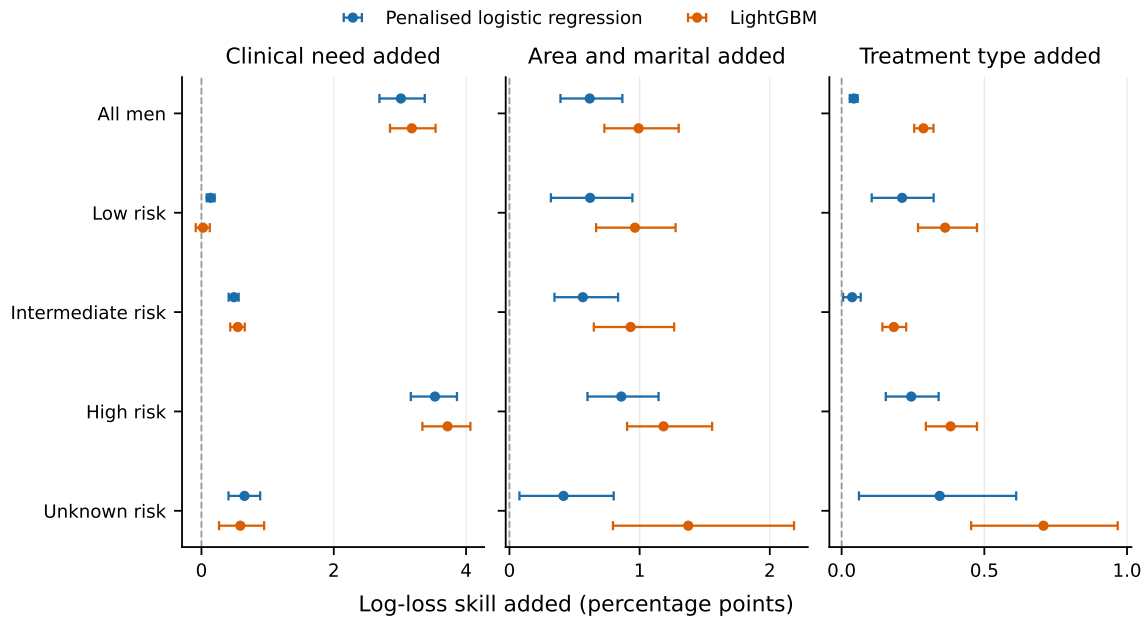


Figure 2. Out-of-fold log-loss skill added by each block of features, in percentage points, by risk group and model type, with 95% cluster bootstrap intervals (500 resamples, no refitting). The dashed line marks no added skill, and the x-axis scale differs between panels.

crement unchanged, and across 10 fold assignments it ranged from 0.61 to 0.62 under logistic regression and 0.98 to 0.99 under LightGBM in all men; the smallest value was above 0 in every stratum and model, as was every interval with county income band clusters. The LightGBM clinical model was weaker than the logistic one in low-risk (0.05 against 0.14 points) and unknown-risk men (0.55 against 0.64), so its larger increment there may partly reflect a weaker reference. Under per-step tuning, logistic regression chose C at the edge of the grid in 8 of 15 stratum and step combinations. Across the ten scenarios the interval for the increment lay above 0 in 98 of 100 combinations, the area rule was met in 9 of 10 scenarios (at 180 days the contrasts were -2.96 and -3.20), and the marital rule in all 10. Between 2.7% and 5.5% of treated men had no recorded interval, depending on rurality; the rurality gap held under extreme bounds (40.2% to 45.7% against 31.3% to 34.0%), and weighting changed group percentages by at most 0.36 points. Among 352,644 intermediate- and high-risk men, 75.4% and 81.1% had a recorded radical prostatectomy or radiotherapy; the area and marital block added 1.96 (1.72 to 2.31) and 2.16 (1.92 to 2.54) points of skill, and never-married men were 6.5 and 5.9 points less likely than married men to have a record.

Discussion

In 330,827 US men with recorded radical prostatectomy or radiotherapy, whether a man waited more than 90 days was only weakly predictable from registry information. Clinical need predicted delay mainly in high-risk men. County rurality, county income and marital status added a small but repeatable increment, stable across tuning, fold assignment, clustering and ten sensitivity analyses. Delay was more common in large metropolitan, higher-income counties and among never-married

men, and it rose over time, in the same direction as national evidence of lengthening intervals [1].

An increment below 1.4 points of log-loss skill improves the average accuracy of individual predictions only slightly, because most variation in waiting is not captured by registry variables. It is nonetheless compatible with standardised differences of 6 to 9 percentage points between specific profiles, because few men sit in the most different profiles; for the cohort as a whole the observed profile differed from the reference by only 1.1 to 1.6 points. The lower delay in high-risk men fits triage, but also hormone therapy ending the interval before radiotherapy: when radiotherapy patients were excluded, high-risk delay was 38.4% rather than 32.8%, and clinical need added 1.66 points in all men rather than 3.01 and 3.18.

The area direction applies to the joint contrast of rurality and income only. It agrees with US studies reporting shorter intervals for non-metropolitan or rural patients [5, 6], but it is not evidence of better rural care. Among high-risk men, recorded curative treatment was less common in non-metropolitan counties not adjacent to a metropolitan area than in large metropolitan counties (75.3% against 81.9%), consistent with a national SEER study of guideline-concordant management [21]; if rural men who would have waited longer are more often managed without recorded treatment, the direction would be exaggerated. Urban men with low-risk disease were less likely than rural men to be managed with surveillance or watchful waiting in an earlier SEER study [22]. Never-married men were both more often delayed and less often recorded as treated; without data on social support or comorbidity the reason cannot be identified, and comorbidity explained only part of a racial gap in definitive treatment in SEER-Medicare [23]. Gradient boosting improved on penalised regression by less than 1 point of skill, and the inference that flexible

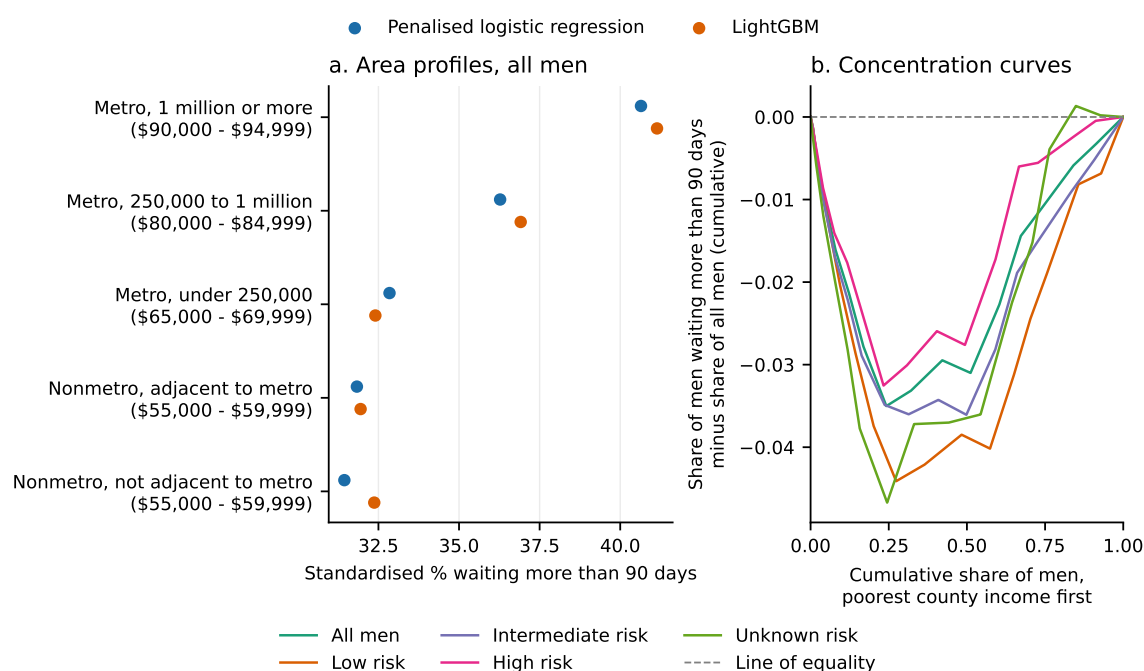


Figure 3. (a) Standardised percentage of all men waiting more than 90 days, with rurality and county income set together to each type of area at the median income band of men living there; each man keeps his own clinical features and year, and marital status is set to married. (b) Crude concentration curves of waiting more than 90 days by county income rank, shown as the difference from the line of equality; values below 0 mean delay is concentrated among men in higher-income counties. Neither panel separates income from rurality.

clinical adjustment did not absorb the signal rests on all men and on intermediate- and high-risk men.

No Australian data were analysed, and the results cannot be compared directly with Australian findings. In Tasmania, men from outer regional and remote areas took a median of 82 days to start active treatment against 75 days in inner regional areas (age-adjusted difference 9.25 days, 95% CI 1.72 to 16.79) [24], and among men not treated with external beam radiotherapy those treated in public facilities started 42 to 59 days later than those treated privately [25]. In Queensland the median treatment interval was 65 days [26], and Tasmanian lung cancer benchmarking found that on average 7% of patients met pathway treatment-time standards [27]. The approach used here, a benchmark outcome with risk-stratified out-of-sample increments and joint area profiles, could be applied to a registry that records remoteness, area disadvantage, treating sector and referral dates; comparing survival between treated and untreated men would additionally require designs that avoid immortal time bias [28].

The limitations are substantial. The export lacked race, insurance, registry and comorbidity, so part of the increment may reflect unmeasured health, insurance or registry practice, and two of the three area and marital variables are county measures. The interval ends at the first treatment of any kind, and diagnosis dates may be recorded inconsistently across registries, settings or years. SEER under-captures treatment [17], summary stage partly uses pathology, and the cohort is conditioned on recorded curative treatment. The main intervals exclude model-fitting variability, standardised contrasts have no intervals, and three amendments were

made after seeing results. The analysis was done by a single author.

Conclusions

Whether a US man with recorded prostatectomy or radiotherapy waits beyond 90 days is mostly unpredictable from registry data. Area and marital characteristics add a small, repeatable increment beyond recorded clinical need, and delay is more common in large metropolitan, higher-income counties among treated men. Registries with referral dates, treating sector and comorbidity could show what this signal represents.

Article information

Author contributions (CRediT): Tanmay Hinge: conceptualisation, methodology, software, formal analysis, data curation, visualisation, writing (original draft, review and editing).

Conflict of interest disclosures: None.

Funding/support: None.

Data sharing statement: SEER data are available from the National Cancer Institute under a Research Data Use Agreement and cannot be shared by the author; aggregate results, code and the protocol are at <https://github.com/tanmayhinge/seer-prostate>.

Ethics: This study analysed only de-identified registry data released by the National Cancer Institute under the SEER Research Data Use Agreement, with no contact with patients and no access to identifying information, so ethics committee approval was not sought. No patients or members of the public were involved in the design or conduct of the study.

Use of AI tools: Code, analyses and manuscript text were developed with assistance from Claude (Anthropic); the author is responsible for the design, checks and interpretation.

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